Software Design Description (SDD) *for*

College Internship Tracking and Management System

Version: 1.0

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1. Introduction

1.1 Purpose

The purpose of this Software Design Description (SDD) is to provide a detailed design of the **Internship Tracking and Management System (ITMS)**. It translates the requirements specified in the SRS into a structured design that developers can implement efficiently.

1.2 Objective

- To design a centralized system for managing internships.
- To define database structure and relationships clearly.
- To ensure efficient handling of students, companies, and applications.
- To support scalability and future enhancements.

1.3 Scope of the Project

The Internship Tracking and Management System (ITMS) is a web-based application designed to streamline and manage internship-related activities within a college. The system allows companies to post internship opportunities, enables students to apply for internships and track their application status, and allows faculty coordinators to monitor student participation and generate reports. It also provides secure authentication with role-based access control and efficiently manages student profiles and internship data.

1.4 Overview of Project

The Internship Tracking and Management System is a web-based application developed to simplify and centralize the management of internship activities. The system enables administrators and companies to manage internship listings, while students can view opportunities, apply for internships, and track their application status. Faculty coordinators can monitor student participation and analyze internship data through reports. The system is built using a relational database (MySQL) and is accessible through a web interface, ensuring efficient data management and easy access for all users.

2. Data Design

2.1 Entity Relationship Diagram

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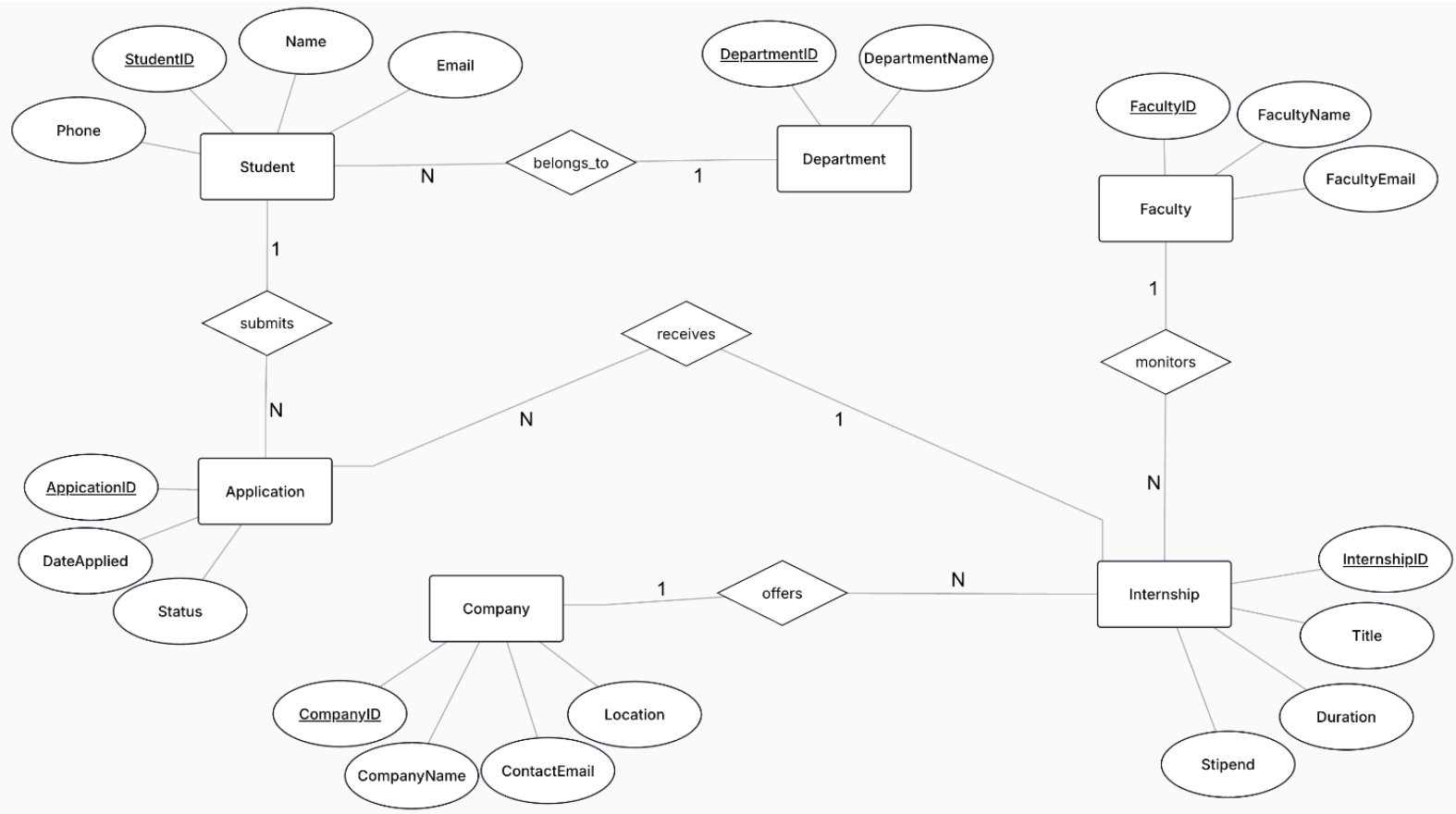
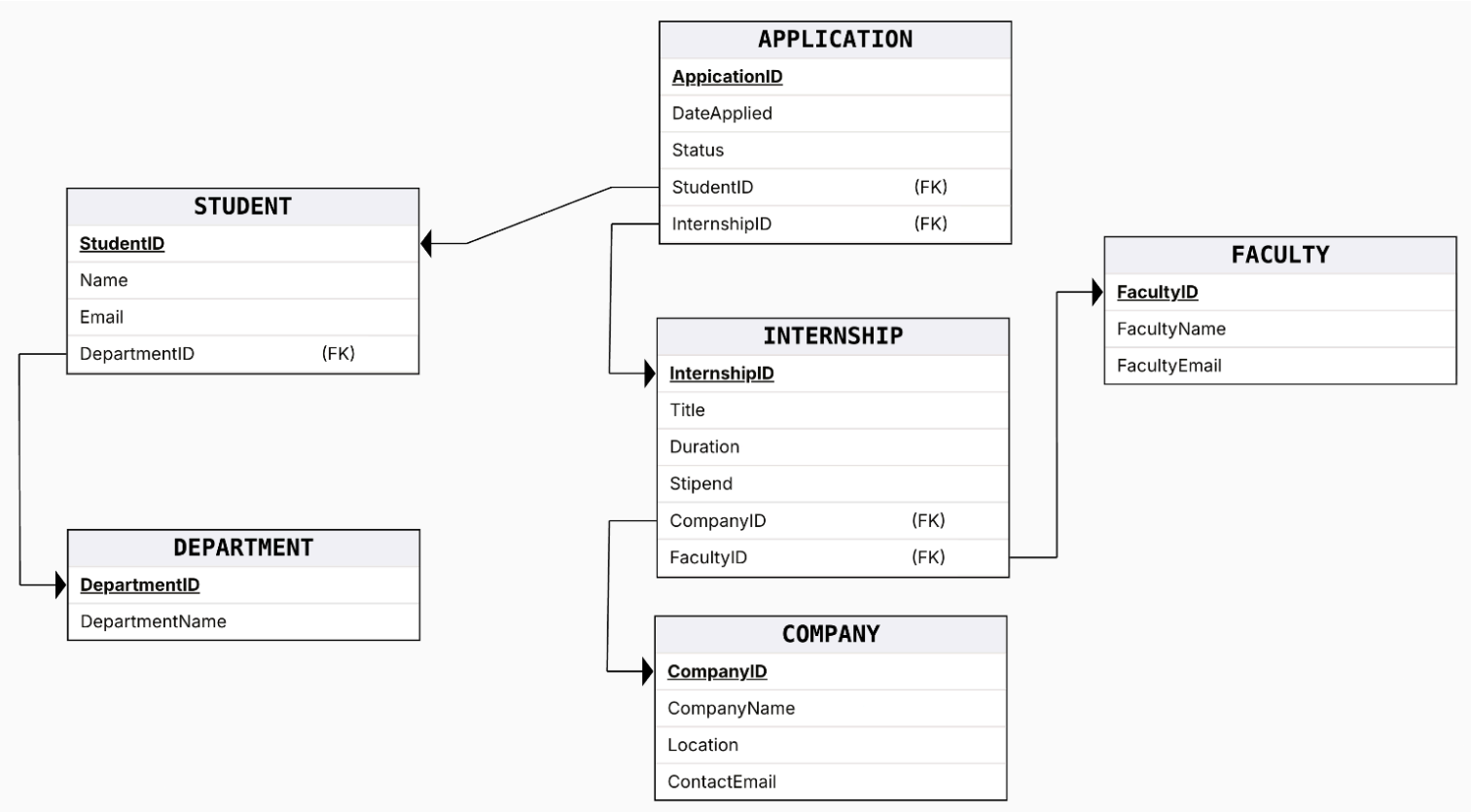


2.2 Conceptual Schema



3. Entities and Attributes

This section describes the **key entities** in the system, their **attributes**, and how they interact.

3.1 Student

Attribute	Type	Key Type
StudentID	INT	Primary Key
Name	VARCHAR	Non-Key
Email	VARCHAR	Non-Key
Phone	VARCHAR	Non-Key
DepartmentID	INT	Foreign Key

3.2 Department

Attribute	Type	Key Type
DepartmentID	INT	Primary Key
DepartmentName	VARCHAR	Non-Key

3.3 Application

Attribute	Type	Key Type
ApplicationID	INT	Primary Key
DateApplied	DATE	Non-Key
Status	VARCHAR	Non-Key
StudentID	INT	Foreign Key
InternshipID	INT	Foreign Key

3.4 Internship

Attribute	Type	Key Type
InternshipID	INT	Primary Key
Title	VARCHAR	Non-Key
Duration	INT	Non-Key
Stipend	DECIMAL(10,2)	Non-Key
CompanyID	INT	Foreign Key
FacultyID	INT	Foreign Key

3.5 Company

Attribute	Type	Key Type
CompanyID	INT	Primary Key
CompanyName	VARCHAR	Non-Key
ContactEmail	VARCHAR	Non-Key
Location	VARCHAR	Non-Key

3.6 Faculty

Attribute	Type	Key Type
FacultyID	INT	Primary Key
FacultyName	VARCHAR	Non-Key
FacultyEmail	VARCHAR	Non-Key

4. System Architecture

The Internship Tracking and Management System (ITMS) follows a **three-tier architecture**, which separates the system into presentation, application, and data layers to ensure modularity, scalability, and maintainability.

4.1 Presentation Layer

This layer represents the user interface of the system. It is accessed through a web browser and provides interaction between users and the system. The interface includes pages such as the login page, dashboard, student profile page, internship listings page, and application tracking page. Users such as students, faculty coordinators, and administrators interact with the system through this layer.

4.2 Application Layer

The application layer contains the core business logic of the system. It processes user requests, enforces rules, and manages workflows such as internship posting, application submission, and status updates. This layer ensures that all operations follow the functional requirements defined in the system, such as maintaining student records, managing internship listings, and tracking application status.

4.3 Data Layer

The data layer is responsible for storing and managing all system data using a relational database (MySQL).It includes tables such as Student, Company, Internship, Application, Faculty, and Department. This layer handles data storage, retrieval, and integrity using SQL queries, ensuring consistency and reliability of the system.

5. Module Design

The Internship Tracking and Management System is divided into several functional modules, each responsible for a specific set of operations derived from the system requirements.

5.1 Student Module

This module manages student-related functionalities. It allows students to create and update their profiles, view available internship opportunities, apply for internships, and track the status of their applications.

5.2 Company Module

This module manages company-related data and operations. It allows administrators to add, update, and manage company details and enables companies to provide internship opportunities.

5.3 Internship Module

This module handles internship listings. It allows authorized users to create, update, and delete internship postings, including details such as role, duration, stipend, and eligibility Criteria.

5.4 Application Module

This module manages internship applications submitted by students. It stores application details, links students with internships, and maintains the application status such as pending, accepted, or rejected.

5.5 Faculty Module

This module enables faculty coordinators to monitor internship activities. Faculty members can view student applications, track participation, and oversee internship records.

5.6 Reporting Module

This module generates reports and analytics related to internship activities. It provides insights such as the number of applications, selected students, and participation statistics across departments or companies.